Dining Room Composition, Fit & Budget Logic

AID-LOG-004 | Revision 1

Purpose. Defines the backend logic used by AIDOOi to translate Dining Room inputs into an appropriate room composition, validate product fit and circulation, apply room-level budget logic, and generate recommendations. Customer journey inputs are defined in AID-SPC-004.

> **Core sequence: Room Dimensions -> Room Size -> Dining Capacity -> Table Fit -> Chair Fit & Circulation -> Remaining Composition -> Budget -> Recommendation.**

1. Core Principles

1. The Dining Table and required seating capacity form the anchor dining set. Remaining products are considered only after the table, chairs and necessary circulation have been validated.

2. Dining Capacity has priority over generic room-size assumptions. AIDOOi must not silently reduce the customer's selected capacity to make a default composition fit.

3. Geometric table fit alone is insufficient. Chair pull-out space and main circulation must also be acceptable.

4. Default Composition is a recommendation baseline, not a hard constraint. Secondary items may be removed when the layout does not support them.

5. Preferred Product Dimensions are guidance, not absolute filters. Final fit must consider room dimensions, room shape, placement zones and circulation.

6. Room Budget Tier is not the same as Product Price Tier. A recommendation may mix product tiers to preserve fit, required capacity and the overall room budget.

7. Style and Colour are matching inputs under AID-DATA-009 and AID-DATA-007. They do not directly define the numeric budget range.

8. Final Composition Price equals the sum of Current Selling Prices of the selected products. If the customer removes products, the total recalculates dynamically and does not need to remain inside the original room-tier range.

2. Dining Room Size Classification

Room Area	Classification
< 8 m²	Compact
8-12 m²	Small
12-18 m²	Medium
18-25 m²	Large
25+ m²	Extra Large

Room Area is system-derived from the Length and Width supplied in AID-SPC-004. The customer does not select this classification.

3. Fit Priority & Product Dependencies

> **Base fit priority: Dining Table -> Required Chairs -> Chair Pull-out -> Main Circulation -> Sideboard -> Rug -> Console -> Lighting -> Decor.**

Dependency	Logic
Chair Quantity	Derived from customer-selected Dining Capacity.

Dependency	Logic
Rug Size	Derived from Table Dimensions plus normal Chair Pull-out requirements; remove the rug if an appropriate rug cannot fit without harming circulation.
Lighting Scale	Derived primarily from Table Dimensions and Table Shape, not room size alone.
Sideboard / Console	Added only after anchor-set fit and circulation are secured.

Capacity Constraint Rule. If the selected capacity cannot be accommodated with acceptable fit and circulation, AIDOOi must surface a Fit Constraint rather than automatically reducing capacity.

4. Composition by Room Size

4.1 Compact Dining Room (< 8 m²)

Product	Qty	Preferred Fit Logic
Dining Table	1	Capacity-driven; generally suited to 2-4 persons when fit allows.
Dining Chairs	Selected capacity	Exact required seating capacity.
Pendant / Ceiling Light	1	Compact; aligned to the table.
Rug	0-1	Small; include only if table, chair movement and circulation remain appropriate.
Sideboard / Cabinet	0	No default; add only when free wall and circulation permit.
Wall Art / Mirror	1	Wall-mounted preferred.
Decor	1-2	Low-volume; table or wall based.

Fit Priority: Table -> Chairs -> Pull-out/Circulation -> Lighting -> Rug -> Decor

4.2 Small Dining Room (8-12 m²)

Product	Qty	Preferred Fit Logic
Dining Table	1	Capacity-driven; commonly 4-6 persons.
Dining Chairs	Selected capacity	Exact required seating capacity.
Pendant / Ceiling Light	1	Scaled to table dimensions and shape.
Rug	1	Small / Medium; derived from table and chair pull-out.
Sideboard / Cabinet	1	Compact, approximately 80-120 cm wide; only after circulation is secured.
Wall Art / Mirror	1	Based on available wall zone.
Decor	2-3	Table, sideboard or wall based.

Fit Priority: Table -> Chairs -> Pull-out/Circulation -> Sideboard -> Rug -> Lighting -> Decor

4.3 Medium Dining Room (12-18 m²)

Product	Qty	Preferred Fit Logic
Dining Table	1	Capacity-driven; commonly 4-6 persons.
Dining Chairs	Selected capacity	Exact required seating capacity.
Pendant / Chandelier	1	Scaled to table dimensions and shape.
Rug	1	Medium / Large; derived from table plus chair pull-out.
Sideboard / Cabinet	1	Approximately 120-180 cm wide.
Mirror / Wall Art	1-2	Based on available wall zones.
Table / Sideboard Decor	2-3	Vase, sculpture, candle holder or equivalent.
Planter	0-1	Only where an appropriate floor/corner zone exists.

Fit Priority: Table -> Chairs -> Pull-out -> Main Circulation -> Sideboard -> Rug -> Lighting -> Decor

4.4 Large Dining Room (18-25 m²)

Product	Qty	Preferred Fit Logic
Dining Table	1	Capacity-driven; commonly 6-8 persons.
Dining Chairs	Selected capacity	Exact required seating capacity.
Pendant / Chandelier	1	Scaled to table dimensions and shape.

Product	Qty	Preferred Fit Logic
Rug	1	Large / XL; derived from table plus chair pull-out.
Sideboard / Cabinet	1	Approximately 160-220 cm wide.
Console Table	1	Approximately 100-160 cm; conditional on fit.
Mirror / Wall Art	1-2	Based on available wall zones.
Table / Surface Decor	3-4	Distributed without overfilling surfaces.
Planter	1	Appropriate floor/corner zone required.

Fit Priority: Table -> Chairs -> Pull-out -> Main Circulation -> Sideboard -> Rug -> Console -> Lighting -> Decor

Large-room restraint: Furniture quantity must not increase merely because the room is larger. Empty space can be part of a good composition.

4.5 Extra Large Dining Room (25+ m²)

Product	Qty	Preferred Fit Logic
Dining Table	1	Capacity-driven; commonly 8-10+ persons.
Dining Chairs	Selected capacity	Exact required seating capacity.
Pendant / Chandelier	1-2	Based on table length, shape and lighting layout.
Rug	1	XL; derived from table plus chair pull-out.
Sideboard / Cabinet	1	Approximately 180-260+ cm wide.
Console Table	1	Approximately 120-180 cm.
Mirror / Wall Art	1-2	Based on available wall zones.
Table / Surface Decor	4-5	Proportionate to available surfaces.
Planter / Floor Decor	1-2	Only in suitable floor zones.

Fit Priority: Table -> Chairs -> Pull-out -> Main Circulation -> Sideboard -> Rug -> Console -> Lighting -> Decor

Secondary dining zone: Do not create a second dining zone by default. A single strong dining composition is preferred unless the layout or project requirement clearly supports another zone.

5. Dining Room Budget Logic

Budget is applied at room-composition level. Room Size defines the expected composition and fit envelope, while Dining Capacity acts as an important cost modifier because it affects both table size and chair quantity.

> **Budget relationship: Room Size + Dining Capacity -> Composition -> Budget Tier -> Style & Colour Matching -> Recommendation.**

The five customer-facing budget tiers are Essential, Value, Standard, Premium and Luxury. Numeric ranges below are target ranges for the complete recommended room composition.

Room Size	Essential	Value	Standard	Premium	Luxury
Compact (<8 m²)	500-1,200	1,200-2,000	2,000-3,500	3,500-6,000	6,000+
Small (8-12 m²)	1,000-2,000	2,000-3,500	3,500-5,500	5,500-9,000	9,000+
Medium (12-18 m²)	1,500-3,000	3,000-5,000	5,000-8,000	8,000-13,000	13,000+
Large (18-25 m²)	2,500-4,500	4,500-7,000	7,000-11,000	11,000-18,000	18,000+
Extra Large (25+ m²)	3,500-6,000	6,000-9,500	9,500-15,000	15,000-25,000	25,000+

Capacity Modifier. Two rooms in the same size class can produce different actual composition prices when their required seating capacities differ. Capacity must be reflected in table selection, chair quantity and budget allocation.

> **Budget Allocation Priority: Dining Table + Required Chairs -> Sideboard -> Rug -> Lighting -> Console -> Decor.**

If the anchor dining set consumes a larger share of the selected budget, AIDOOi may use lower Product Price Tiers for secondary items. It must not reduce required capacity or compromise fit merely to preserve decorative completeness.

Product-level prices and tier logic reference AID-LOG-001. Dining Chair prices are normalized per chair and Dining Table prices are Table Only, excluding chairs.

6. Recommendation Rules

1. Validate anchor-set fit before selecting secondary products.
2. Respect the customer's selected Dining Capacity unless a Fit Constraint must be surfaced.
3. Use Style and Colour preferences to rank products that already satisfy fit, availability and budget logic.
4. Allow product-tier mixing inside a room budget tier.
5. Use Current Selling Price when calculating the final composition total.
6. Recalculate the total dynamically when the customer removes products from the proposed composition.
7. Do not overfill larger dining rooms solely to consume space or budget.

7. Standard References

Reference	Standard
AID-SPC-004	Dining Room Path Standard
AID-DATA-007	Product Colour Standard
AID-DATA-009	Product Style Standard
AID-LOG-001	Product Price Range Standard
AID-CAT-001	Seating Standard
AID-CAT-003	Tables Standard
AID-CAT-005	Storage & Casegoods Standard
AID-CAT-006	Lighting Standard
AID-CAT-007	Rugs & Soft Furnishings Standard

Reference	Standard
AID-CAT-008	Decor & Accessories Standard

Revision 1 status: Approved MVP logic. Numeric thresholds and composition preferences may be revised later using observed product inventory, user behavior and recommendation performance.